

Video Transcript

Video 5: Classes and Objects in Java (05:06)

If you're not familiar with classes and objects, and that wouldn't be surprising today, in fact, a lot of languages like JavaScript and Python take you a really long way without using them. And in fact many development platforms have removed them due to the additional friction and not really a lot of help that they can provide. But as you can imagine, because of what I've mentioned when it comes to the language of Java, that it would be deep at the heart of it and in fact it is. And so because of it, let's look at some of the basics and use here a couple of classes and bring them together.

So, when it comes to the overall template, you can see here that we could define a class as a thing. And there used to be in the old days deep hierarchies of these that were constructed. But keeping it simple, you can imagine here a classical thing and inside of it, it would have two main constructs, one of them being the public part, that is the methods, as you can see there, the functions that can define what operations can be done on this type of object and then some private ones that would be limited to be visible within the class.

And so in general you could construct the thing. Once you define this thing, you would be able to create instances of it or occurrences of it, create new ones, and then within it you would have some functions, some methods that would be defined and that would provide support to be able to carry out the classes work. So, if we took a look at a couple of these and created an example, you can see here that I have created a Person program and it's a classical person and inside of it, I have a constructor which is a method inside that we use to initialize the class. And in this case, we want to be able to pass in a parameter to set the name of a person.

And then what we have what follows next, there is a get and a set method that would allow us to perform those operations, that is get a name or set a name. And separately from that, you have a consumer file that is going to consume that class. So, as you can see there, within the method inside of consumer, we have there a new instance being created of Person. And once we have that new instance, then we can invoke some of the methods inside of it, the ones that are public. And you can see there how I am invoking the name.

So, let's go ahead and write this up and we'll start off by creating the file. And in this case, the file is going to be `Person.java`. Inside of it, I'm going to go ahead and paste our code, and as I scroll up, you can see there the the class being defined. You can see there our private variable, our constructor, our method that we used to define or to instantiate the class and you can see the other two supporting methods. So, I'm going to go ahead and exit from here, save the file. Then I'm going to go ahead and compile it just to make sure that, that does in fact compile.

And as you can see there, that has been created. Now I'm going to create the `Consumer` class, and my file is going to be `Consumer.java`. And now I'm going to go ahead and paste the code and I'm going to go ahead and remove some of the unnecessary spacing there. And as you can see there, it simply instantiates a class `Person`, and then we call one of its methods. So, I'm going to go ahead and save, and then I'm going to go ahead and compile. And as you can see there that has now compiled. Now let's go ahead and run the `Consumer` program that consumes the `Person` class. And now I'm going to go ahead and hit 'return', and as you can see there, that instantiated the class. Let's set the name as Peter Parker, and then we were able to recall it. Now if we took a look at the file system, you can see there that I have both of the files we created and we also have the classes that were generated when we compiled each of those files. So, now it's your turn. Make sure you can replicate what I have done, make sure that you can create some new ones.

Just think of some scenarios, think of something that you want to have multiple of, create your own program. And one last note here, note that all of these files are inside of the same directory. So, in order for the path resolution in this case, for the class to be discovered, it needs to be given the way that I have written it. It needs to be in the same directory.